

Mridul Mahajan

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EDUCATION

Boston University

Ph.D. Student in Data Sciences, GPA: 3.86/4.0, in progress

Advisor: Prof. Aldo Pacchiano

Boston, US

Sept. 2024 – Present

Indian Institute of Information Technology (IIIT) Allahabad

B.Tech. and M.Tech. in Information Technology, GPA: 9.61/10.0

Prayagraj, India

2017 – 2022

PUBLICATIONS

- [5] **Mahajan M.**, Pacchiano A.†, Zhang X.† “Learning to Undo: Transfer Reinforcement Learning under State Space Transformations”, *under review at NeurIPS 2026*
- [4] Schneider J., **Mahajan M.**, Chen L., Guist S., Schölkopf B., Posner I., Büchler D. “Sim-to-Real Transfer for Muscle-Actuated Robots via Generalized Actuator Networks”, CoRL 2026 [paper]
- [3] Hottung A., **Mahajan M.**, Tierney K. “PolyNet: Learning Diverse Solution Strategies for Neural Combinatorial Optimization”, ICLR 2025 [paper]
- [2] **Mahajan M.**, Tzannetos G., Radanovic G.†, Singla A.†. “Learning Embeddings for Sequential Tasks Using Population of Agents”, IJCAI 2024 [paper]
- [1] Chandra R.*, Wang X.*, **Mahajan M.**, Kala R., Palugulla R., Naidu C., Jain A., Manocha D. “METEOR: A Dense, Heterogeneous, and Unstructured Traffic Dataset With Rare Behaviors”, ICRA 2023 [paper]

* equal contribution, † equal advising

WORK & RESEARCH EXPERIENCE

Graduate Research Assistant

Boston University, Advisor: Prof. Aldo Pacchiano

Boston, US

Sept. 2024 – Present

- **Transfer Reinforcement Learning under State Space Transformations.** Proposed LUMOS, an algorithm that learns an undo map by matching state-feature statistics and transfers a source policy without target rewards. Proved improved sample efficiency over tabula rasa RL under linearly- Q^* realizable MDPs.
- **Learning Stratified Policies for Inference-Time Objectives.** Developing a population-policy framework that directly optimizes inference-time objectives such as $\text{pass}@k$ by learning complementary strategies.

Research Assistant

Bielefeld University, Advisor: Prof. Kevin Tierney

Bielefeld, Germany

Aug. 2023 – July 2024

- Proposed PolyNet, an approach that learns a diverse and complementary set of solution strategies for neural combinatorial optimization. PolyNet achieves state-of-the-art performance on several routing problems.

Guest Researcher

MPI-IS, Advisors: Prof. Dieter Büchler and Prof. Bernhard Schölkopf

Tübingen, Germany

Feb. 2023 – July 2024

- Developed a sim-to-real framework for robots actuated by pneumatic artificial muscles. The framework learns a predictive model of actuator dynamics from real-robot trajectory data.

Visiting Scholar

Max Planck Institute for Software Systems, Advisor: Prof. Adish Singla

Saarbrücken, Germany

Jan. 2022 – Nov. 2022

- Designed an information-theoretic framework to learn embeddings for RL tasks using a diverse population of agents. The embeddings support agent-performance prediction and task selection based on desired characteristics.

SERVICE AND OUTREACH

Reviewer

NeurIPS (2024, 2025), ICLR (2025, 2026), ICML (2026), AISTATS (2026)

AWARDS & SCHOLARSHIPS

Institute Gold Medal

Awarded for the highest GPA in the graduating batch.

IIIT Allahabad

2023

Pre-doctoral Scholarship

Awarded to support research at MPI-IS.

Max Planck Institute for Intelligent Systems

2023

Research Fellowship

Awarded to support research at MPI-SWS.

Max Planck Institute for Software Systems

2022

WISE Scholarship

Awarded to support a research internship in Germany.

German Academic Exchange Service (DAAD)

2021

Merit Incentive Award

Awarded for excellent academic performance.

IIIT Allahabad

2017 – 2019